

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)
ON
MULTILINGUAL SMS SCAM DETECTION
submitted in partial fulfilment of the requirements for the award of degree of
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In
COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Monalika– 2210990582

Payal Dhiman – 2210990647

Prachi Malik– 2210990661

Supervised By:

Dr. Monika Aggarwal

Associate Professor

**Chitkara University Institute
of Engineering & Technology**



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**MULTILINGUAL SMS SCAM DETECTION**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Monika Aggarwal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 28-April-2026

Monalika – 2210990582

Payal Dhima – 2210990647

Prachi Malik – 2210990661

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Monika Aggarwal

Associate Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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The computational work and experiments were carried out using the Google Colab platform with the help of Python libraries such as TensorFlow, Keras, and scikit-learn, which greatly facilitated the implementation of the proposed model.

We also acknowledge the use of publicly available SMS datasets from Kaggle, which were further extended and customized for Hinglish SMS classification, forming the foundation of our research.

Finally, we would like to thank all contributors and team members involved in this project for their cooperation and efforts in successfully completing this work.

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ABSTRACT

This project focuses on detecting SMS-based scams, particularly in Hinglish (Hindi + English mixed language), which is commonly used in India. Traditional spam detection systems fail to identify such messages due to language mixing and informal structure.

To solve this problem, a hybrid machine learning and deep learning model is proposed. The system combines TF-IDF (Term Frequency-Inverse Document Frequency) for keyword-based feature extraction and LSTM (Long Short-Term Memory) for understanding contextual patterns in messages.

The model uses a late fusion technique, where both statistical and sequential features are merged to improve detection accuracy. Unlike traditional binary classification (spam vs ham), this system classifies SMS messages into 12 different scam categories such as OTP fraud, bank scams, job fraud, and lottery scams.

The proposed model achieves an accuracy of 92.47%, making it highly effective for real-time scam detection. This system helps improve mobile security and provides users with better awareness of different types of fraud.

INTRODUCTION

With the rapid growth of mobile communication in India, SMS-based scams (smishing) have become a major cybersecurity issue. Attackers now use **Hinglish language**, mixing Hindi and English words, to bypass traditional spam filters that are designed mainly for single-language texts.

Existing systems often fail because they rely either on keyword matching or simple machine learning models trained on English datasets. These systems cannot effectively detect **code-mixed and informal messages**, which are common in real-world scenarios.

To address this gap, this project proposes a **hybrid late fusion model** that combines both machine learning and deep learning approaches. The TF-IDF model captures important scam-related keywords, while the LSTM model understands the sequence and context of the message.

Another key improvement is moving beyond binary classification. Instead of simply labeling messages as spam or not spam, this system classifies them into **multiple scam categories**, providing more detailed and actionable insights.

This approach improves detection accuracy and helps users identify the exact type of threat, making the system more practical and useful.

METHODOLOGY

The methodology of this project is based on a **hybrid dual-stream architecture**, combining statistical and deep learning techniques for better performance.

Step 1: Dataset Preparation

A custom Hinglish dataset is created and expanded using data augmentation techniques. The dataset includes SMS messages categorized into **12 different classes** such as banking fraud, OTP scams, and promotional messages.

Step 2: Data Preprocessing

- Removal of missing values
- Cleaning text (whitespace, noise removal)
- Tokenization and encoding
- Conversion of labels into numerical format using one-hot encoding

Step 3: Feature Extraction (Two Streams)

1. TF-IDF (Statistical Stream):

Extracts important keywords and assigns weights based on their frequency.

2. LSTM (Contextual Stream):

Processes sequences of words to understand the meaning and intent behind messages.

Step 4: Hybrid Late Fusion

Outputs from both TF-IDF and LSTM models are combined into a single feature vector. This improves detection by considering both keyword importance and contextual meaning.

Step 5: Classification

A Softmax layer is used to classify messages into **12 categories**, instead of just spam/ham.

Step 6: Evaluation

The model is evaluated using metrics like:

- Accuracy
- Precision
- Recall
- F1-score

The final system achieves **92.47% accuracy**, showing strong performance.

TOOLS AND TECHNOLOGIES

This project uses a combination of machine learning, deep learning, and web-based tools to build an efficient SMS scam detection system.

Programming Language

- **Python** is used as the primary language due to its strong support for machine learning and data processing.

Libraries and Frameworks

- **NumPy & Pandas** → for data handling and preprocessing
- **Scikit-learn** → for TF-IDF feature extraction and basic ML utilities
- **TensorFlow / Keras** → for building and training the LSTM model
- **Matplotlib / Seaborn** → for visualization (graphs, confusion matrix)

Machine Learning Techniques

- **TF-IDF (Term Frequency-Inverse Document Frequency)** for extracting important keywords from SMS
- **LSTM (Long Short-Term Memory)** for understanding sequence and context of messages

Other Technologies

- **Jupyter Notebook / Google Colab** → for development and testing
- **Dataset Source** → Kaggle SMS dataset (extended and customized for Hinglish)

These tools together help in building a robust hybrid model capable of handling multilingual and code-mixed text.

IMPLEMENTATION

The implementation of the system is divided into multiple stages:

Step 1: Data Collection

A Hinglish SMS dataset is created by combining existing datasets and manually adding code-mixed messages. The dataset is expanded using **data augmentation techniques** to improve model performance.

Step 2: Data Preprocessing

- Cleaning text data (removing spaces, noise)
- Handling missing values
- Tokenizing text into words
- Padding sequences for equal length
- Encoding labels into numerical format

Step 3: TF-IDF Model

- Extracts keyword-based features
- Converts text into numerical vectors
- Helps detect common scam words like “lottery”, “KYC”, “OTP”

Step 4: LSTM Model

- Embedding layer converts words into dense vectors
- LSTM layer processes sequences to capture context
- Learns patterns in Hinglish sentence structure

Step 5: Hybrid Model (Late Fusion)

- Outputs from TF-IDF and LSTM are combined
- A dense layer processes the merged features
- Dropout is used to prevent overfitting

Step 6: Classification

- Final layer uses **Softmax activation**
- Classifies messages into **12 categories**

Step 7: Testing

- Model is tested using unseen data
- Performance is evaluated using accuracy and other metrics

MAJOR FINDINGS/RESULTS

The results of the project show that the hybrid approach significantly improves SMS scam detection.

Key Results

- Achieved **92.47% accuracy**
- Better performance compared to:
 - Traditional ML models (~84%)
 - Deep Learning models (~82%)

Observations

- TF-IDF effectively detects **important scam keywords**
- LSTM successfully understands **context and intent**
- Combining both reduces **false positives and false negatives**

Category-wise Performance

- High accuracy in major categories like:
 - Promotional messages
 - Conversations
- Strong detection of scam types like:
 - Bank fraud
 - OTP scams
 - Job scams

Confusion Matrix Insight

- Most predictions are correct (strong diagonal values)
- Some minor misclassification occurs in similar categories

Conclusion from Results

The hybrid model performs better than individual models because it captures both:

- **What is written (keywords)**
- **How it is written (context)**

CONCLUSION

This project successfully addresses the growing problem of SMS-based scams, especially in Hinglish language, which is commonly used in India. Traditional spam detection systems are not effective for such code-mixed text, leading to security risks for users.

The proposed **hybrid model combining TF-IDF and LSTM** overcomes these limitations by capturing both keyword importance and contextual meaning. The use of **late fusion architecture** ensures that the strengths of both machine learning and deep learning techniques are utilized effectively.

Unlike traditional systems that only classify messages as spam or not, this system provides **multi-class classification into 12 categories**, giving more detailed insights into the type of scam.

With an accuracy of **92.47%**, the model proves to be highly efficient and reliable for real-time SMS scam detection. Overall, this system enhances user security and contributes to improving cybersecurity solutions in multilingual environments.

FUTURE SCOPE

Although the system performs well, there are several areas for improvement:

- **Real-time Mobile Integration**

The model can be integrated into mobile applications to provide live SMS scam detection.

- **Use of Advanced Models**

Future work can include advanced NLP models like **BERT** for better contextual understanding.

- **Larger Dataset**

Expanding the dataset with more real-world Hinglish messages can improve accuracy further.

- **Multi-language Support**

Extend the system to support other regional languages like Punjabi, Tamil, etc.

- **Explainable AI**

Adding explanations for predictions can help users understand why a message is flagged as a scam.

REFERENCES

1. Al-Zahrani, A., & Al-Ayyoub, M. (2020). Hybrid CNN-LSTM Model for SMS Spam Detection in Arabic and English.
2. Guna, M. G., Karthikeyan, S., et al. (2020). Deep Learning to Filter SMS Spam.
3. Al-Zoubi, A. M., Mora, A. M., & Faris, H. (2021). Multilingual Spam Detection using Word Embeddings and SVM.
4. Ramnujam, E., Shankar, K., & Sharma, A. (2022). Multi-lingual SMS Spam Detection using Hybrid Techniques.
5. Gupta, A., Kumar, P., & Singh, N. (2023). Hinglish NLP: Code-Mixed SMS Classification.
6. Khan, R., et al. (2023). Hybrid TF-IDF and Bi-LSTM for Text Classification.
7. Gupta, S., & Gharde, S. (2024). Phishing SMS Detection in Indian Languages.
8. Altunay & Albayrak (2025). Deep Learning Based SMS Spam Detection.
9. Srivastava, R., & Singh, A. (2025). Contextual Analysis of Code-Mixed Text using BERT.
10. Sharma, S., & Mehta, R. (2026). Multilingual Financial Fraud Detection using ML.
11. Kaggle Dataset: SMS Spam Collection Dataset